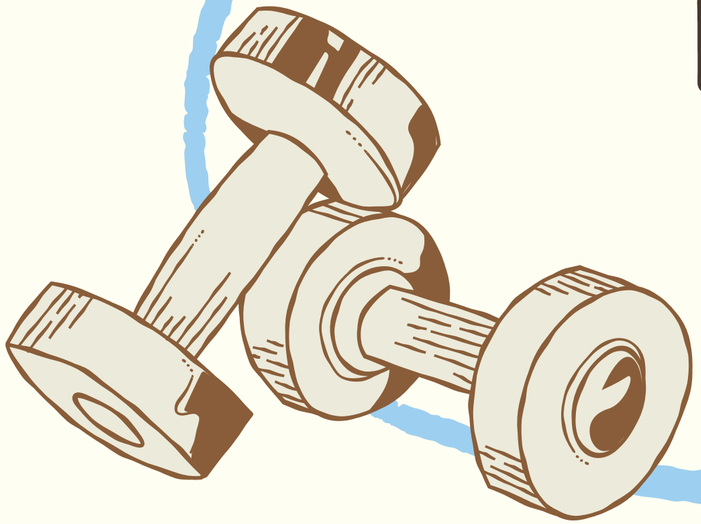
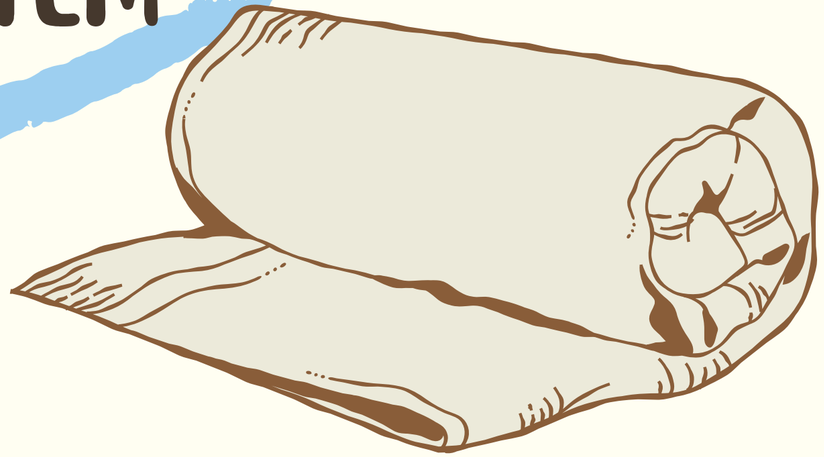
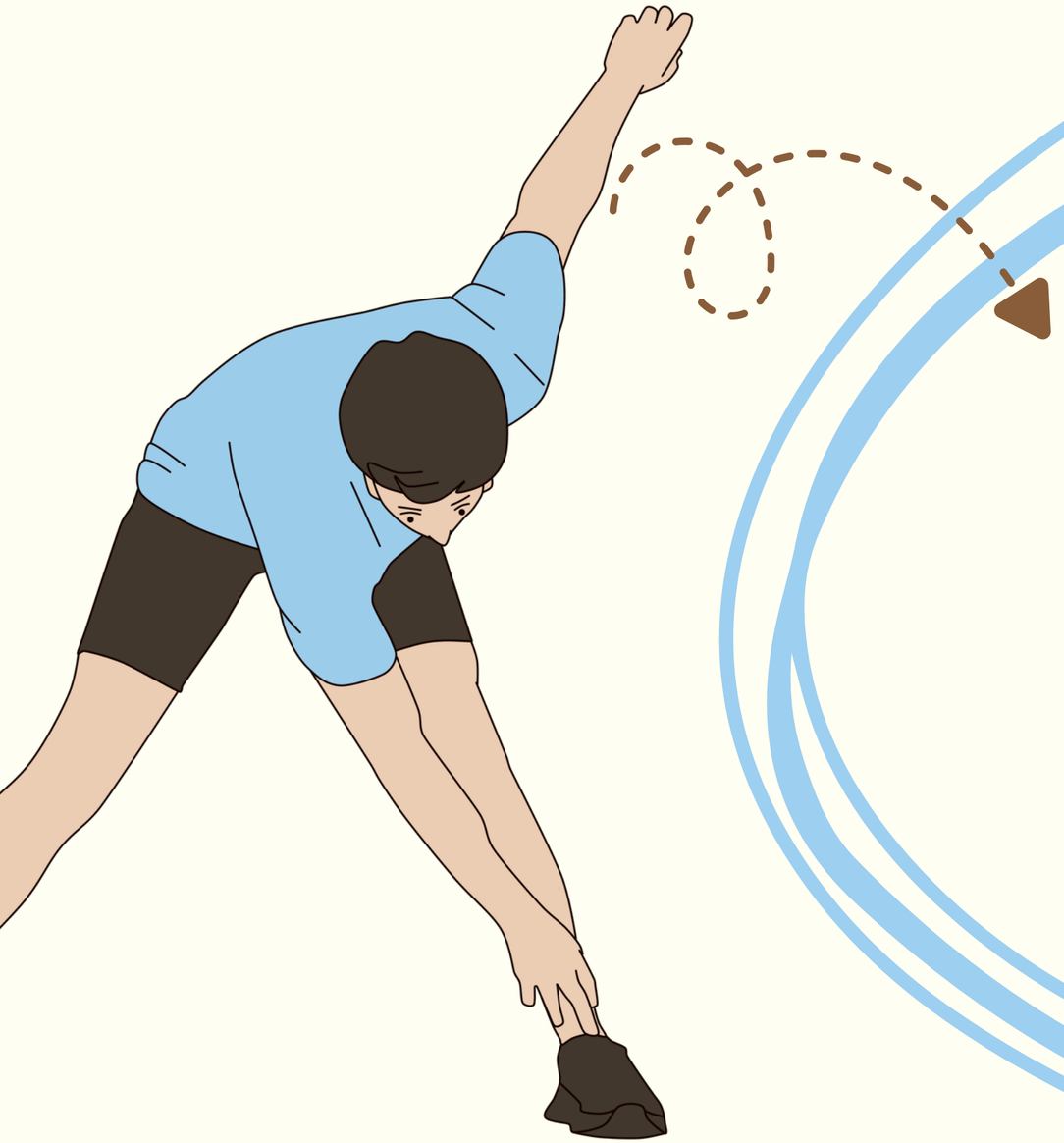
DBMS REVIEW-1

GYM MEMBERSHIP AND FITNESS SESSION MANAGEMENT SYSTEM









INTRODUCTION

The system maintains information about:

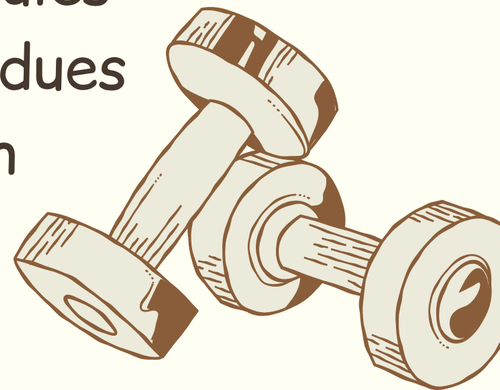
- Members
- Trainers
- Membership Plans
- Fitness Sessions
- Session Schedules
- Bookings
- Attendance
- Payments and Feedback



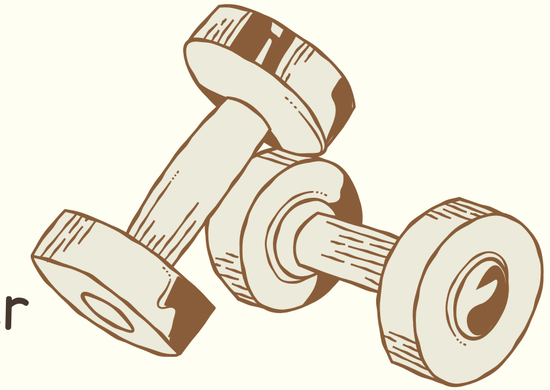
PROBLEM IDENTIFICATION

In many gyms, membership and session information is managed using registers, spreadsheets or separate records. This can lead to:

- Difficulty managing large numbers of members
- Duplicate or incorrect membership records
- Difficulty checking session availability
- Difficulty tracking attendance
- Difficulty managing trainer schedules
- Difficulty tracking payments and dues
- Time-consuming report generation



OBJECTIVES



1. Maintain a centralized database of member records
2. Manage trainer and fitness session information
3. Manage membership plans and gym branches efficiently
4. Track fitness session schedules
5. Record member bookings for sessions
6. Track attendance for each session
7. Maintain records of membership payments
8. Collect member feedback on sessions
9. Reduce duplicate and incorrect data
10. Improve the efficiency of gym managemen



SCOPE OF THE PROJECT

1. Member Management
- Member details
- Membership plans
- Payments
- Branch details

2. Trainer Management
- Trainer details
- Specialization
- Fitness sessions
- Equipment

3. Session Management
- Session schedules
- Bookings
- Attendance
- Workout plans

4. Reporting
- Active members
- Session-wise attendance
- Revenue summary
- Feedback reports

LITERATURE SURVEY

Existing Approach	Limitations
Manual registers	Time-consuming
Excel spreadsheets	Data duplication and inconsistency
Separate member and trainer records	Difficult to integrate
Manual attendance marking	Possibility of errors
Manual payment tracking	Delayed and error-prone
Manual report generation	Takes more time

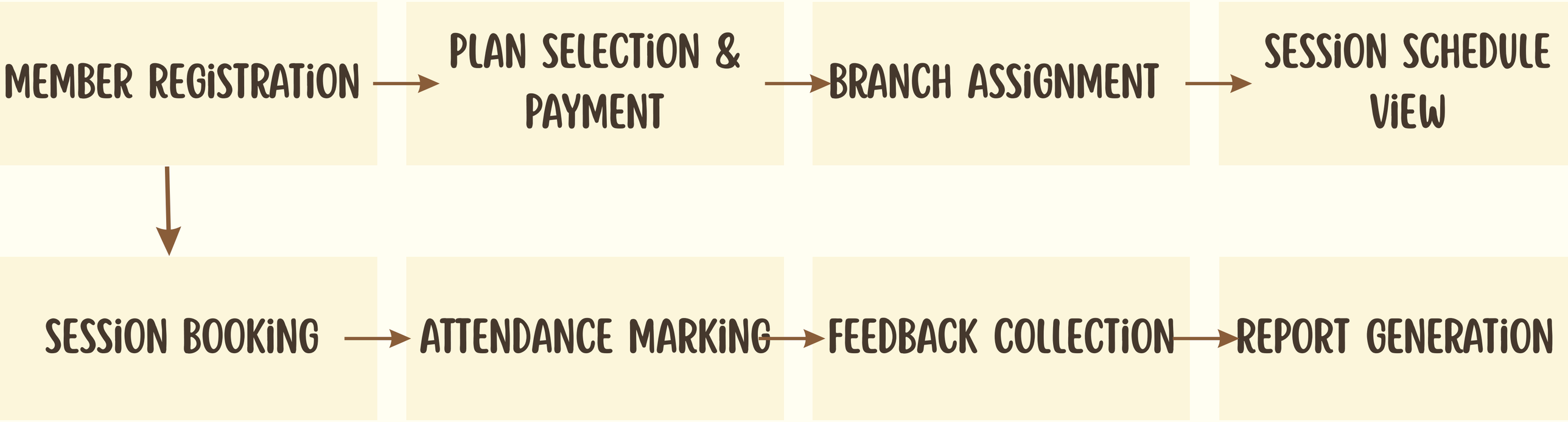
PROPOSED SYSTEM PROVIDES

- Centralized data
- Automated attendance tracking
- Structured relationships
- Faster searching
- Better data integrity
- Automated reports

CONCLUSION

Existing approaches can manage basic gym information, but a centralized relational database provides better organization, consistency and efficient retrieval of gym data.

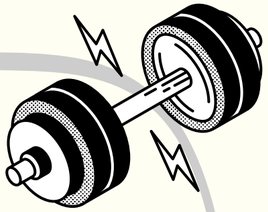
PROPOSED SYSTEM (WORKFLOW)



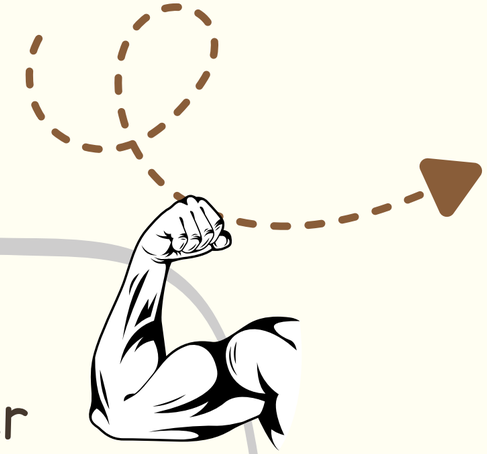
USERS OF THE SYSTEM



1. Gym Administrator
   - Manage members
   - Manage trainers
   - Manage membership plans
   - Create fitness sessions
   - Generate reports

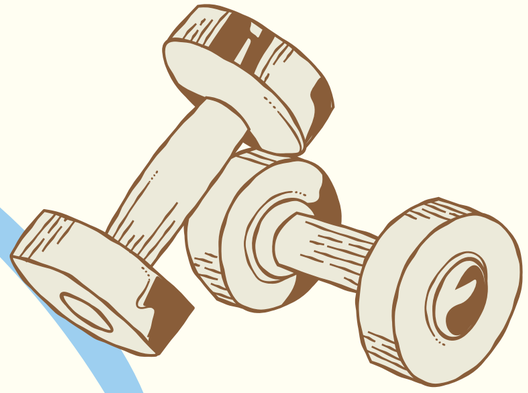


2. Member
   - View available sessions
   - Book sessions
   - Make payments
   - View attendance
   - Give feedback
   - View membership status



3. Trainer
   - Conduct sessions
   - Mark attendance
   - Create workout plans
   - View bookings
   - Update session details
   - View member feedback

FUNCTIONAL REQUIREMENTS



- 1-2: Manage Members & Trainers
- 3: Manage Membership Plans
- 4: Create Sessions & Schedules
- 5-6: Check Availability & Book Sessions
- 7: Track Attendance
- 8: Record Membership Payments
- 9: Create Member Workout Plans
- 10: Collect Session Feedback
- 11: Search Membership & Sessions
- 12: Generate Reports

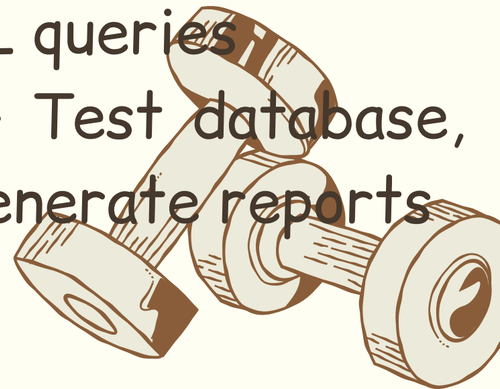


MAIN ENTITIES

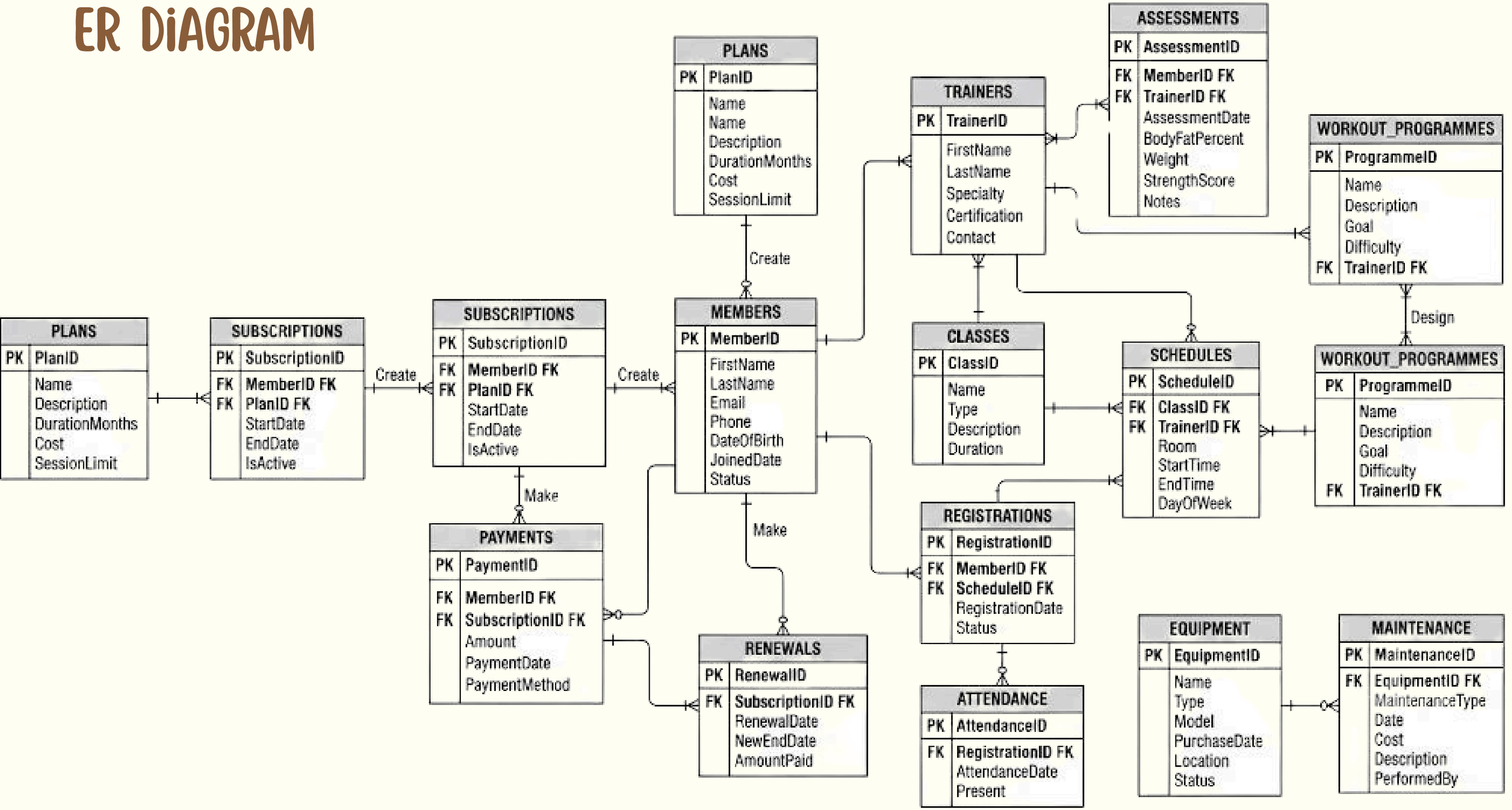
Member Group: Member, Gym Branch, Membership Plan, Payment
Session Group: Trainer, Equipment, Fitness Session, Session Schedule
Transaction Group: Booking, Attendance, Feedback, Workout Plan

PROJECT PLANNING

- Requirement Analysis - Identify requirements, Identify users, Define project scope
- Database Design - Identify entities, ER diagram, Functional dependencies
- Database Implementation - Relational schema, Create tables, Normalization
- ApplicationDevelopment-Develop application,ImplementCRUD operations, Write SQL queries
- Testing & Reporting - Test database, Insert sample data, Generate reports



ER DiAGRAM


PLANS
PK PlanID
Name
Description
DurationMonths
Cost
SessionLimit
Create
SUBSCRIPTIONS
PK SubscriptionID
FK MemberID FK
FK PlanID FK
StartDate
EndDate
IsActive
Create
MEMBERS
PK MemberID
FirstName
LastName
Email
Phone
DateOfBirth
JoinedDate
Status
Make
PAYMENTS
PK PaymentID
FK MemberID FK
FK SubscriptionID FK
Amount
PaymentDate
PaymentMethod
Make
RENEWALS
PK RenewalID
FK SubscriptionID FK
RenewalDate
NewEndDate
AmountPaid
PLANS
PK PlanID
Name
Description
DurationMonths
Cost
SessionLimit
Create
TRAINERS
PK TrainerID
FirstName
LastName
Specialty
Certification
Contact
ASSESSMENTS
PK AssessmentID
FK MemberID FK
FK TrainerID FK
AssessmentDate
BodyFatPercent
Weight
StrengthScore
Notes
WORKOUT_PROGRAMMES
PK ProgrammeID
Name
Description
Goal
Difficulty
FK TrainerID FK
Design
WORKOUT_PROGRAMMES
PK ProgrammeID
Name
Description
Goal
Difficulty
FK TrainerID FK
CLASSES
PK ClassID
Name
Type
Description
Duration
SCHEDULES
PK ScheduleID
FK ClassID FK
FK TrainerID FK
Room
StartTime
EndTime
DayOfWeek
REGISTRATIONS
PK RegistrationID
FK MemberID FK
FK ScheduleID FK
RegistrationDate
Status
ATTENDANCE
PK AttendanceID
FK RegistrationID FK
AttendanceDate
Present
EQUIPMENT
PK EquipmentID
Name
Type
Model
PurchaseDate
Location
Status
MAINTENANCE
PK MaintenanceID
FK EquipmentID FK
MaintenanceType
Date
Cost
Description
PerformedBy

INITIAL RELATIONAL SCHEMA

MEMBER

Member_ID (PK)
Branch_ID (FK)
Name, Email, Phone
Join_Date

TRAINER

Trainer_ID (PK)
Branch_ID (FK)
Name, Specialization
Experience_Years

BOOKING

Booking_ID (PK)
Member_ID (FK)
Schedule_ID (FK)
Booking_Date, Status

GYM_BRANCH

Branch_ID (PK)
Branch_Name
Location
Contact_Number

EQUIPMENT

Equipment_ID (PK)
Branch_ID (FK)
Equipment_Name
Category, Quantity

ATTENDANCE

Attendance_ID (PK)
Booking_ID (FK)

Gym_Membership_Fitness_Session_DBMS.pptx

MEMBERSHIP_PLAN

Plan_ID (PK)
Plan_Name
Duration_Months
Price

FITNESS_SESSION

Session_ID (PK)
Trainer_ID (FK)
Session_Name, Category
Duration_Minutes

FEEDBACK

Feedback_ID (PK)
Member_ID (FK)
Session_ID (FK)
Rating, Comments

PAYMENT

Payment_ID (PK)
Member_ID (FK)
Plan_ID (FK)
Amount, Payment_Date

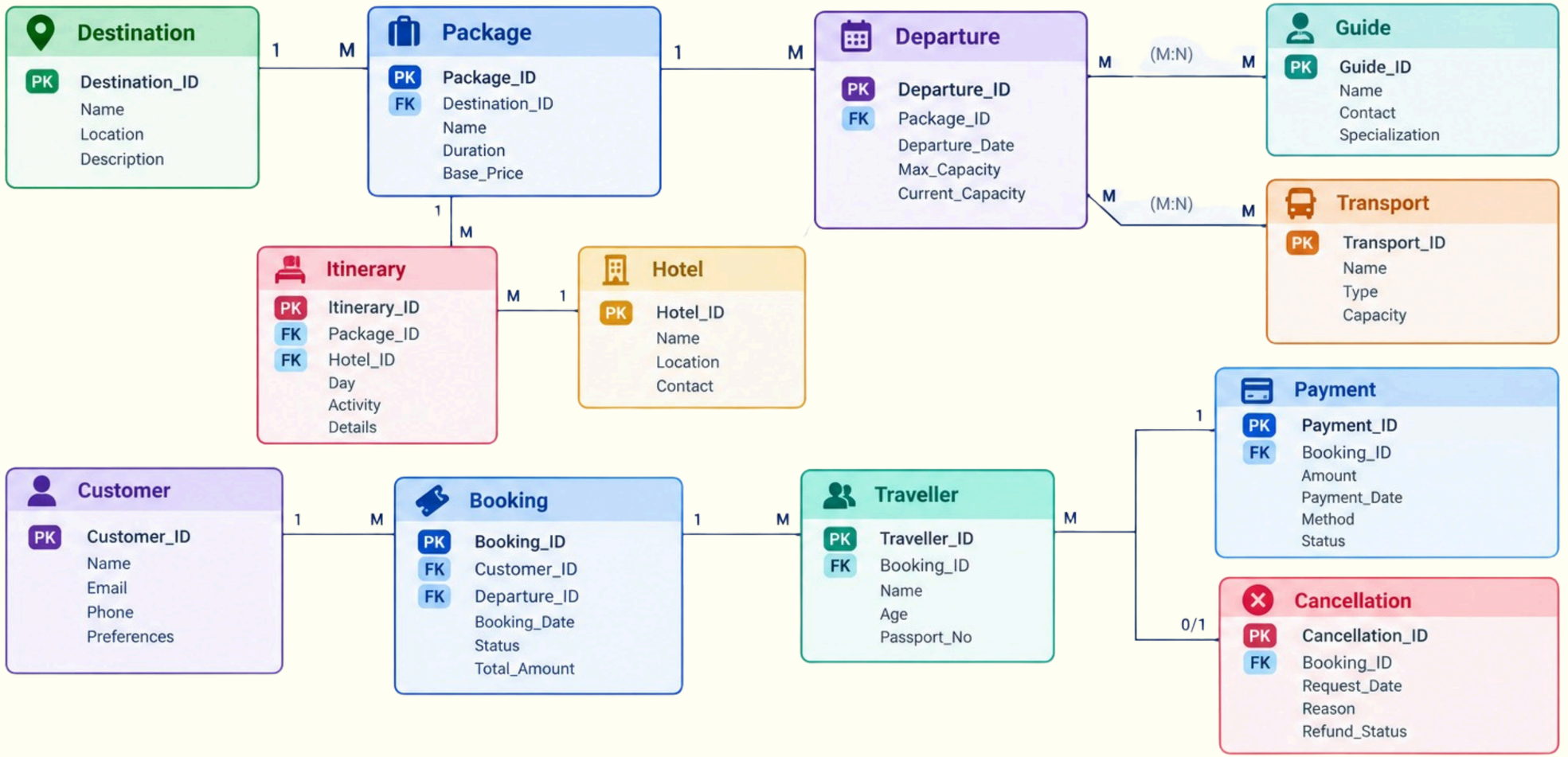
SESSION_SCHEDULE

Schedule_ID (PK)
Session_ID (FK)
Session_Date, Time_Slot
Room_No

WORKOUT_PLAN

Workout_ID (PK)
Trainer_ID (FK)
Member_ID (FK)
Goal, Duration_Weeks

PRIMARY KEYS & RELATIONSHIPS



Primary Keys (PK)

Customer_ID, Package_ID, Destination_ID, Itinerary_ID, Departure_ID, Hotel_ID, Guide_ID, Transport_ID, Booking_ID, Traveller_ID, Payment_ID, Cancellation_ID

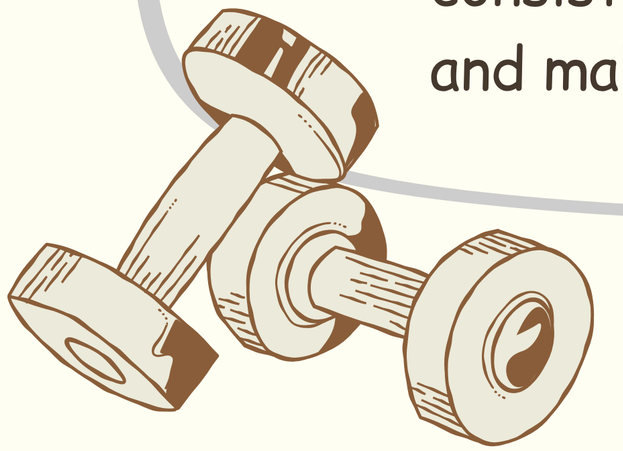
Key Relationships

- Customer → Booking → Traveller / Payment / Cancellation
- Destination → Package → Itinerary / Departure
- Itinerary → Hotel
- Departure ↔ Guide / Transport

PK = Primary Key
FK = Foreign Key
1 = One
M = Many
0/1 = Zero or One

CONCLUSIONS

A centralized, structured system for gym membership management.The Gym Membership and Fitness Session Management System reduces manual work, improves data consistency, helps track attendance and payments accurately, and makes gym information easier to manage and retrieve.



THANK YOU



